SLIDE 1 PROBLEM & WHY IT MATTERS

Seniors are growing faster than the clinics that serve them

Team name

[Team name]

Problem statement

A3 HealthPulse: community healthcare demand forecasting

The problem in one sentence

Some towns are running out of clinic capacity faster than others, and **nobody can see which ones first,** or whether the new polyclinics will open in time.

Who is affected, and how badly

Singapore's 65+ population grew **45% between 2015 and 2022**, to 735,000. Polyclinics and GP clinics grew 28% over the same years.

Why it matters now

Healthier SG has signed up 1.4 million residents (May 2026), so more care lands on GPs. MOH is choosing polyclinic sites right now, and **Bishan's has already slipped past 2030.**

Seniors per clinic in Singapore

293

in 2022, up from **259 in 2015**



Change from 2015 to 2022, from the DAISI data pack (population by age; MOH primary care facilities).

And the pipeline is already slipping

2025 plan		2026 update
Bishan polyclinic by 2030	→	**Pushed beyond 2030**

SLIDE 2 SOLUTION & DATA

CareRunway tells planners where the next clinic should go

Your solution in one sentence

A planning tool that uses **machine learning** to forecast seniors per clinic in every area to 2031, checks MOH's announced polyclinics, and ranks where one new clinic helps most.

Who uses it, and what decision it changes

Planners at the three regional health clusters, **deciding where the next Healthier SG clinic goes.**

Datasets we will use

1. Residents by area, age and sex, 2000 to 2025 (SingStat)
2. Clinic and polyclinic locations (data.gov.sg, MOH)
3. Hospital admissions 1987 to 2026 (DAISI pack); admission rates by age (MOH)

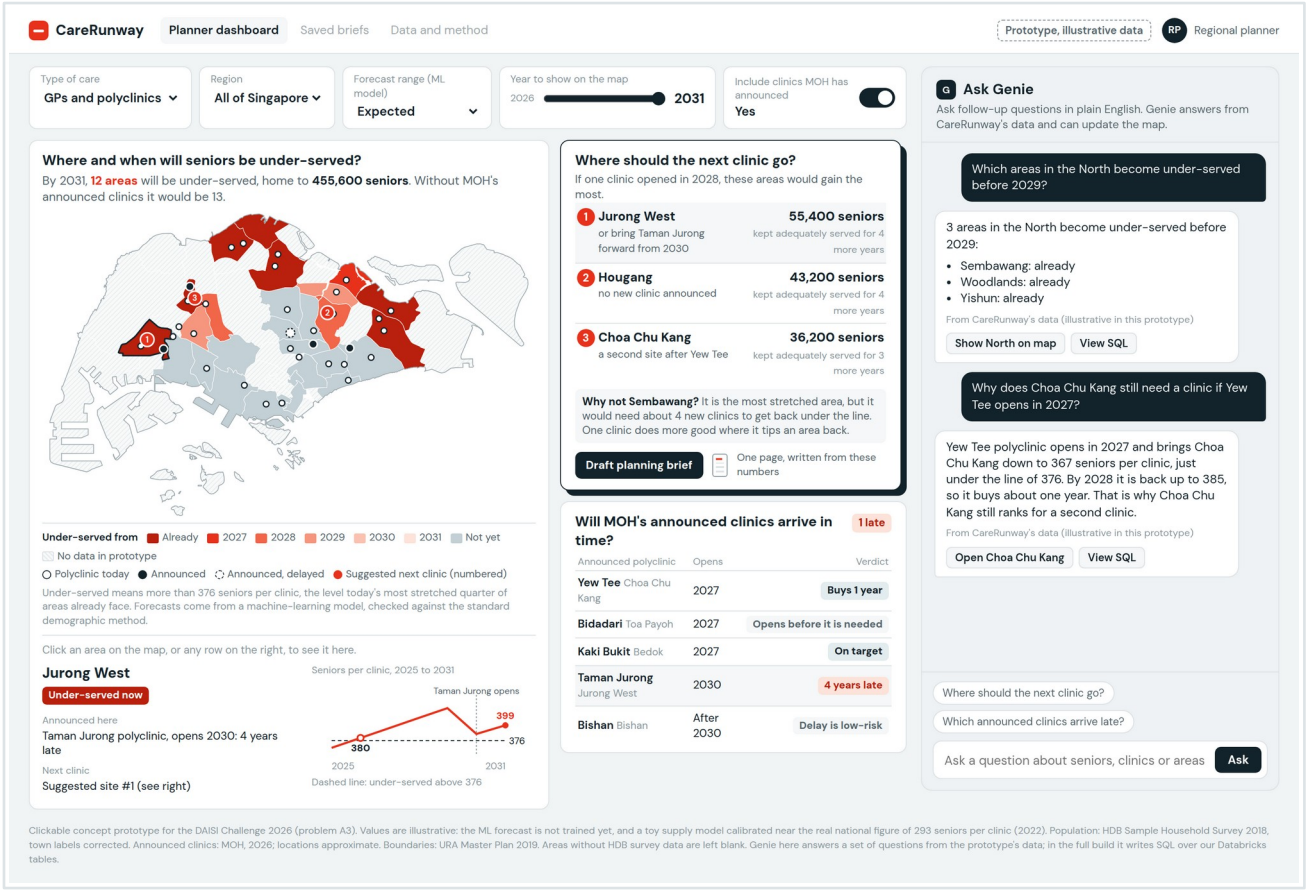
While cleaning the data, **we found 8 towns mislabelled** in a government dataset. It overturned one of our own early findings.

Different from a generic dashboard

Most dashboards show where seniors live today. Ours says **where the next clinic should go,** checks MOH's own plans, and tests its forecast against the standard demographic method.

Try the prototype: claude.ai/artifact/KWxnv7oRJRx4aS9JTt8oXy

Concept only: illustrative data, and many features are not built yet.



SLIDE 3 DATABRICKS ARCHITECTURE & IMPACT

Our data is already in Databricks. Here is how it becomes a recommendation.

Sources
SingStat, data.gov.sg, MOH, DAISI pack, URA boundaries

→

Ingestion
Lakeflow Jobs land raw files in a Unity Catalog Volume

→

Transform
Delta bronze, silver, gold: label fixes, geocoding, catchments

→

Models
MLflow: ML area forecast, tested against the standard method; admissions forecast

→

Outputs
Site optimiser, AI/BI map dashboard, Genie space, App for briefs

We use the ML forecast only where it wins: **train to 2015, predict 2020; train to 2020, predict 2025,** and compare its error with the standard method.

Paste a Databricks screenshot here

Seniors per clinic, 2009 to 2022, queried from workspace.carerunway

What we'll demo

A planner flips planned clinics on and off, clicks an area, asks Genie a question, and leaves with **the top three sites and a one-page brief.**

How we'll measure impact

Seniors living in under-served areas by 2031, with and without our recommended sites, and forecast error versus the standard method.

What we can finish in two weeks

Week 1: pipeline, standard forecast, **ML forecast and backtest.** Week 2: site optimiser, planned-clinics check, dashboard, Genie. Stretch: nursing homes.

Team: [Name] (data and models), [Name] (supply and mapping), [Name] (product and pitch)